EXPERIMENT – 2

Stages of Compilation and Execution of a C Program

A) Aim

To study and demonstrate the different stages of compilation of a C program using the -save-temps option and to execute a C program for calculating the area of a circle.

B) Requirements

- Linux environment (Kali Linux / CoCalc terminal).
- GCC (GNU C Compiler).
- A C source file named area.c.
- Terminal for compiling and executing the program.

C) Theory

A C program passes through several stages before it is executed. The main stages are preprocessing, compilation, assembly and linking.

The GCC compiler can preserve the intermediate files by using the -save-temps option. For a source file area.c, the command below keeps the intermediate files generated during compilation:

```
gcc -save-temps area.c -o area
```

The important files produced are:

- area.c – C source code written by the programmer.
- area.i – Preprocessed C source file. Header files are expanded and preprocessing directives are handled.
- area.s – Assembly language code generated from the C program.
- area.o – Object file containing machine-level code that is not yet linked.
- area – Final executable program produced after linking.

D) C Program to Calculate Area of a Circle

Formula used:

```
Area = π × r × r
```

The following program takes the radius as input and calculates the area of the circle.

```
#include <stdio.h>

int main()
{
    float radius, area;

    printf("Enter radius of circle: ");
    scanf("%f", &radius);
```

```
    area = 3.14159 * radius * radius;

    printf("Area of circle = %.2f\n", area);

    return 0;
}
```

E) Procedure and Compilation Stages

Step 1: Create the C source file

Create a file named area.c and enter the C program given above.

```
nano area.c
```

Save the file after entering the program.
area.c containing the C program.

```c
#include<stdio.h>
#define PI 3.14
int main(void)
{
    int radius;
    float area;
    printf("Enter the radius: ");
    scanf("%d", &radius);
    area = PI*radius*radius;
    printf("Area = %f\n", area);
    return 0;
}
```

Step 2: Compile using -save-temps

Use the following command. The -save-temps option tells GCC to keep the intermediate files.

```
gcc -save-temps area.c -o area
```

After successful compilation, GCC keeps the intermediate files such as area.i, area.s and area.o, along with the final executable area.

```
ls
```

Show the compilation command and the generated files using ls.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ nano area.c

┌──(kali㉿kali)-[~/Desktop]
└─$ cc -save-temps area.c

┌──(kali㉿kali)-[~/Desktop]
└─$ ls
a-area.i  a-area.o  a-area.s  a.out  area.c

┌──(kali㉿kali)-[~/Desktop]
└─$
```

Step 3: Preprocessing stage – area.i

In preprocessing, header files are expanded and preprocessing directives such as #include are processed. The result is stored in area.i.

```
gcc -E area.c -o area.i
```

To view the beginning of the preprocessed file:

```
head area.i
```

```
Show the area.i file or its output using head.
```

```
┌──(kali㉿kali)-[~/Desktop]
└─$ head a-area.i
# 0 "area.c"
# 0 "<built-in>"
# 0 "<command-line>"
# 1 "/usr/include/stdc-predef.h" 1 3 4
# 0 "<command-line>" 2
# 1 "area.c"
# 1 "/usr/include/stdio.h" 1 3 4
# 28 "/usr/include/stdio.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 1 3 4
# 33 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 3 4

┌──(kali㉿kali)-[~/Desktop]
└─$ less a-area.i

┌──(kali㉿kali)-[~/Desktop]
└─$ head a-area.i
# 0 "area.c"
# 0 "<built-in>"
# 0 "<command-line>"
# 1 "/usr/include/stdc-predef.h" 1 3 4
# 0 "<command-line>" 2
# 1 "area.c"
# 1 "/usr/include/stdio.h" 1 3 4
# 28 "/usr/include/stdio.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 1 3 4
# 33 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 3 4

┌──(kali㉿kali)-[~/Desktop]
└─$
```

Step 4: Compilation stage – area.s

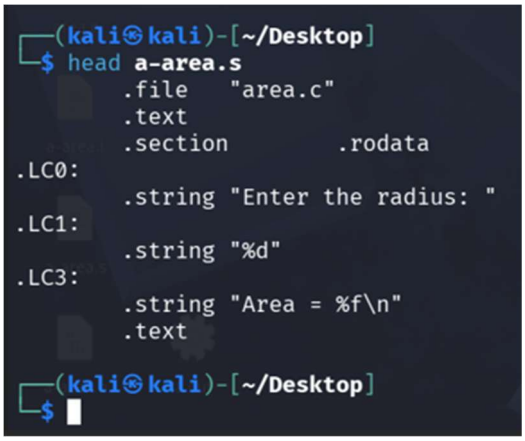
The compiler converts the preprocessed C program into assembly language instructions. The assembly code is stored in area.s.

```
gcc -S area.c -o area.s
```

To view the assembly file:

```
head area.s
```

Show area.s containing assembly instructions.



Step 5: Assembly stage – area.o

The assembler converts the assembly language code into machine-level object code. The output is stored in the object file area.o.

```
gcc -c area.s -o area.o
```

The file command can be used to identify the object file:

```
file area.o
```

Show the creation/identification of area.o using the terminal.

Step 6: Linking stage – final executable

The linker combines the object file with the required libraries and produces the final executable program.

```
gcc area.o -o area
```

Check the generated executable:

```
ls -l area
```

Show the linking command and the final executable file named area.

F) Execution of the C Program

Run the final executable using the following command:

```
./area
```

Example input and output:

```
Enter radius of circle: 5
Area of circle = 78.54
```

For radius = 5, the calculated area is approximately 78.54 square units.

Show the execution of ./area, the entered radius and the calculated area.

G) Summary of Compilation Stages

Stage	File / Command	Purpose
Source	area.c	Original C program written by the programmer.
Preprocessing	area.i / gcc -E	Processes headers and preprocessing directives.
Compilation	area.s / gcc -S	Converts C code into assembly language.
Assembly	area.o / gcc -c	Converts assembly into object code.
Linking	area / gcc area.o -o area	Links object code with libraries to create the executable.

H) Result

The stages of compilation of a C program were studied successfully using GCC and the -save-temps option. The intermediate files area.i, area.s and area.o were generated, the final executable was created, and the program was executed successfully to calculate the area of a circle.